Does a mixed model keep a plant GWAS calibrated? A re-analysis of Atwell et al. (2010) with 8 times more SNPs

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1 Question

Atwell et al. (2010) ran genome-wide association studies (GWAS) on 107 phenotypes in 199 *Arabidopsis thaliana* inbred lines. They report that a simple test (Wilcoxon) is badly confounded by population structure, that the EMMA mixed model removes the excess of association for flowering-related traits, and also that EMMA's p-values "are not always well estimated and should be interpreted with caution". This report re-asks three narrow questions with a different implementation and a much denser SNP set:

1. Does the pattern replicate: strong inflation without correction, near-null calibration with a mixed model?
2. Does a leave-one-chromosome-out (LOCO) kinship, which avoids using a SNP's own region to model relatedness (Yang et al., 2014), change calibration or hits compared with an all-SNP kinship?
3. Do four well-known loci, chosen in advance, localise with the denser SNPs: *FLC*, *FRI* and *FT* for flowering traits, and *DOG1* for seed dormancy?

This is not an attempt to "beat" the paper, which already used a mixed model; it is a check on calibration and localisation with different data and software.

2 Data

Phenotypes. AraPheno study 1 (the Atwell et al. phenotypes) (Togninalli et al., 2020), 107 phenotypes with 51 to 135 accessions each (median 92) after matching to genotypes. Replicate rows were averaged per accession, and each phenotype was rank-based inverse-normal transformed so that a parametric test is comparable with the paper's rank-based Wilcoxon test.

Genotypes. The 1001 Genomes imputed binary SNP matrix (1001 Genomes Consortium, 2016): 10,709,949 SNPs for 1,135 accessions. **Only 137 of the 199 lines in the original paper are in this release**, so these analyses have less power than the original. SNPs with minor-allele frequency below 5% among those 137 accessions were dropped, leaving 1,826,193 SNPs, about 8 times the 216,130 SNPs in the original study. I found no licence statement for the genotype matrix, so the repository contains download scripts, not data.

3 Methods

All code is in `src/`; the statistical core (`src/lmm.py`) has tests (`tests/`) that check it against `scipy`'s OLS and against a brute-force generalised-least-squares computation.

- **OLS:** simple linear regression of the transformed phenotype on each SNP, no correction.
- **Mixed model (all-SNP kinship), EMMAX-style (Kang et al., 2010):** $y = \mu + x\beta + g + e$ with $g \sim N(0, \sigma_g^2 K)$. K is a realised relationship matrix from every 20th SNP (about 91,000 SNPs). The variance ratio $\delta = \sigma_g^2/\sigma_e^2$ is estimated once per phenotype by REML under the null and then fixed for all SNPs (the "P3D" approximation), after which each SNP needs a closed-form weighted regression.
- **Mixed model with LOCO kinship:** the same, but each chromosome is scanned with a kinship computed from SNPs on the other four chromosomes.

- **Significance:** Bonferroni, $0.05/1{,}826{,}193 = 2.7 \times 10^{-8}$. This is conservative because nearby SNPs are in strong linkage disequilibrium (LD).
- **Calibration:** genomic inflation factor λ (median χ^2 over 0.455) per phenotype and method.
- **Candidate loci, fixed before looking at results:** for each pre-specified trait–gene pair (12 flowering traits against *FLC*, *FRI*, *FT*; 2 dormancy traits against *DOG1*; 38 pairs) I take the smallest p-value within the gene plus or minus 50 kb. Because that window holds about a thousand SNPs, a small minimum is unremarkable, so each is compared with 2,000 random windows of the same width: the **window empirical p** is the share of random windows with an equal or smaller minimum.

Software. Python 3.12, NumPy, SciPy, h5py; a fixed random seed for the null windows.

4 Results

4.1 Calibration: the paper's qualitative finding replicates

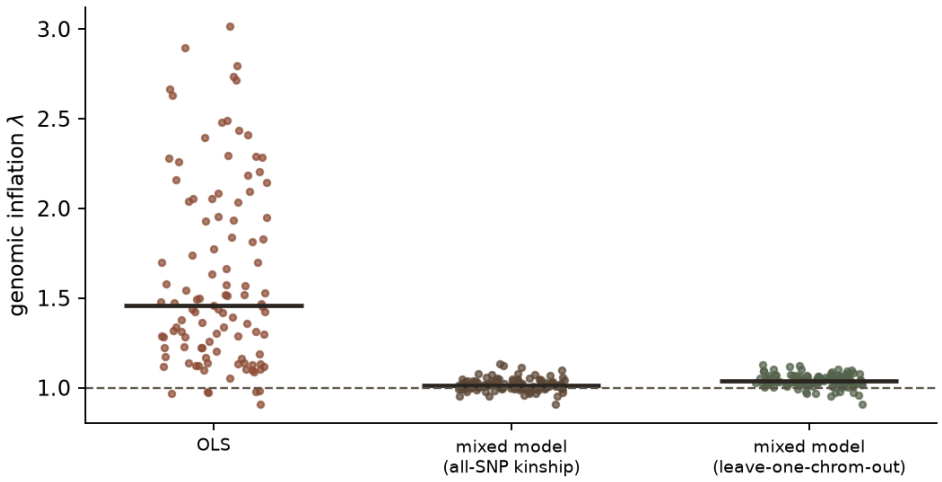


Figure 1: Genomic inflation factor for each of the 107 phenotypes under three methods. Bars: medians. Dashed line: $\lambda = 1$ (well calibrated).

Without correction, 76% of phenotypes have $\lambda > 1.2$ (median 1.46, interquartile range 1.21 to 2.00, maximum 3.02). With a mixed model the inflation is gone: median 1.01 with the all-SNP kinship and 1.04 with LOCO kinship, and no phenotype exceeds $\lambda = 1.13$ (Figure 1). This matches what Atwell et al. (2010) describe for flowering-related traits, here across all phenotypes and with a different implementation.

4.2 What this does to the number of "hits"

Table 1: Bonferroni-significant SNPs ($p < 2.7 \times 10^{-8}$) across the 107 phenotypes. SNP counts overstate independent signals because neighbouring SNPs are in LD; the last column is a rough count of independent regions.

Method	Phenotypes with ≥ 1 hit	Hit SNPs	Distinct 200 kb regions
OLS	52	48,631	2,156
Mixed model, all-SNP kinship	15	2,314	212
Mixed model, LOCO kinship	16	2,260	211

37 phenotypes have at least one OLS hit and no LOCO mixed-model hit; only 1 has the reverse. Flowering time

shows it clearly: for FT16, OLS gives $\lambda = 3.02$ and 10,737 significant SNPs spread across all five chromosomes, while the LOCO mixed model gives a single one (Figure 2).

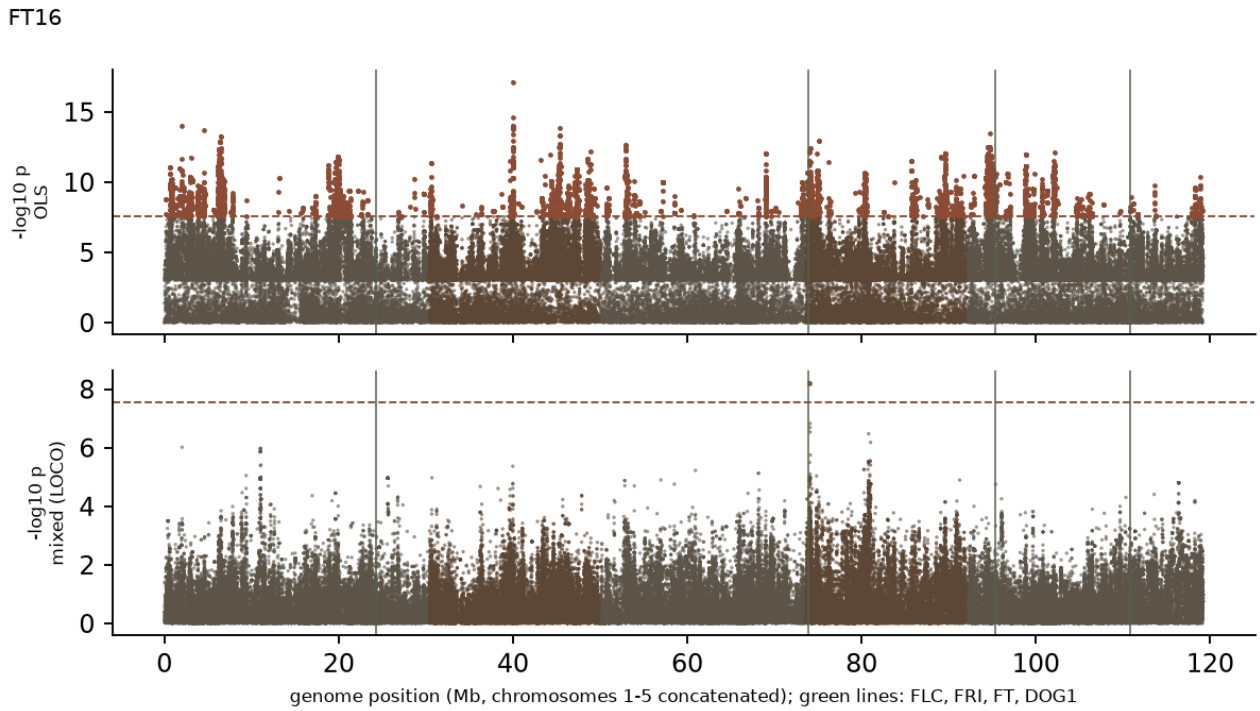


Figure 2: FT16 (n = 134), unadjusted (top) and mixed model with LOCO kinship (bottom). Dashed line: Bonferroni threshold. Green lines: FLC, FRI, FT, DOG1.

The single mixed-model hit for FT16 lies at chromosome 4, position 435,252. It looks adjacent to *FRI* on the concatenated axis, but *FRI* ends at 271,503, so the hit is about 164 kb away and outside the pre-specified window. I do not count it as localising on *FRI*.

Phenotypes with the most mixed-model hits are not flowering traits and have small samples (for example YEL, n = 64, 1,792 LOCO hit SNPs). I did not investigate them; many such SNPs are likely to be in LD with one or a few lead variants.

4.3 All-SNP versus leave-one-chromosome-out kinship

The two mixed models give almost the same hits (2,314 against 2,260 SNPs, 212 against 211 regions). Calibration differs slightly: LOCO has the higher median λ (1.04 against 1.01). A paired Wilcoxon test makes that difference highly detectable ($p = 1.2 \times 10^{-17}$) but it is small in size. It is consistent with the all-SNP kinship slightly deflating test statistics because it includes SNPs near the one being tested, which LOCO is designed to avoid (Yang et al., 2014); I did not test that mechanism directly. For these data the choice does not change the conclusions.

4.4 Do the known loci localise?

Table 2: Pre-specified trait-locus pairs with window empirical p < 0.05 under the LOCO mixed model.

Trait	Locus	OLS	Mixed (all-SNP)	Mixed (LOCO)
FT10	FLC	0.038	0.001	0.003
FT Field	FT	0.412	0.021	0.016
0W GH FT	FT	0.511	0.014	0.018
FT16	FLC	0.066	0.011	0.019

Trait	Locus	OLS	Mixed (all-SNP)	Mixed (LOCO)
LN22	FLC	0.100	0.042	0.030
LN22	FRI	0.080	0.035	0.037
SDV	FLC	0.288	0.051	0.040
LN16	FT	0.653	0.035	0.041

Of the 38 pre-specified pairs, 8 have a window empirical $p < 0.05$ under the LOCO mixed model, against about 1.9 expected by chance; under OLS only 2 do. Four of the eight are flowering traits at *FLC* (FT10, FT16, LN22, SDV), three are at *FT* (0W GH FT, FT Field, LN16) and one is at *FRI* (LN22, which also appears at *FLC*); FT22 at *FLC* just misses under LOCO (0.053); none involves *DOG1* (seed-dormancy windows: empirical p from 0.71 to 0.89), and the *FLC* expression phenotype does not localise on *FLC* or *FRI* (empirical p 0.39 and 0.17).

Two cautions apply. First, the flowering phenotypes are strongly correlated with each other (Atwell et al., 2010), so these eight pairs are not eight independent confirmations. Second, after a Bonferroni correction over 38 pairs (threshold 0.001), only FT10–*FLC* under the all-SNP kinship ($p = 0.001$, borderline) would survive; under LOCO it is 0.003, which does not. The evidence is suggestive for *FLC* and *FT* in flowering traits, and absent for *DOG1* at this sample size.

4.5 Heritability estimates

The REML estimates of $\sigma_g^2/(\sigma_g^2 + \sigma_e^2)$ have a median of 0.72, but 18% of phenotypes sit at the upper boundary (> 0.99) and 5% at the lower (< 0.01). With 51 to 135 lines these estimates are imprecise (my own simulation test gave a standard deviation of about 0.3 at $n = 200$), so I treat them only as an input to the mixed model, not as findings.

5 Discussion and limitations

What this shows. On 1.83 million SNPs and 137 of the original lines, an independent mixed-model implementation reproduces the paper's calibration result across all 107 phenotypes, and a leave-one-chromosome-out kinship changes little. Mixed models also recover more flowering-gene signals than uncorrected tests (8 against 2 pairs below 0.05), though the evidence is modest.

What it does not show. It does not improve on the paper's conclusions, and I did not compare my hit lists with the paper's supplementary tables, so I cannot say whether the same loci are found. The data overlap the original study (the same phenotypes and 137 of the same lines), so this is a robustness check, not an independent replication.

- **Power.** Only 137 of 199 lines are in the genotype release; sample sizes are 51 to 135. A non-significant result, such as for *DOG1*, is not evidence of no effect.
- **Method choices.** Phenotypes were rank-inverse-normal transformed; the paper's choices may differ. The P3D approximation fixes the variance ratio per phenotype. Kinship uses about 91,000 thinned SNPs. Bonferroni correction over all SNPs is conservative under strong LD.
- **Genotypes.** The matrix is imputed, and imputation error is not modelled. SNPs were filtered at 5% minor-allele frequency among the 137 lines.
- **Candidate-gene test.** Pre-specified, but the pairs are correlated and the window (gene plus or minus 50 kb) differs from the paper's 20 kb windows around its top SNPs.
- **Data.** No licence statement was found for the 1001 Genomes genotype matrix, so it is not redistributed here; `src/00_download.sh` fetches it.

6 Provenance and reproducibility

All data are public (AraPheno, the 1001 Genomes, Ensembl Plants for gene coordinates). Run `src/00_download.sh`, then `src/01_prepare.py`, `src/02_scan.py` (about 40 minutes on a laptop) and `src/03_summarise.py`, then render this report. Tests: `micromamba run -n atwell python -m pytest` (4 tests: OLS against `scipy`, the rotated GLS against a brute-force computation, REML recovery of a known

variance ratio, and that the mixed model controls structure OLS does not). The code and report were written and run with the assistance of Claude Code (Anthropic); I am responsible for checking and understanding the results. No part derives from university coursework or assessed material.

References

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